Forecasting Next-Day PM2.5 in Dublin Using Mobile Air Quality, Weather, and Traffic Data

Jakob Lutkemeier, Rosalie LaCroix, Ellen Beattie, Roudra Sarkar

Trinity College Dublin

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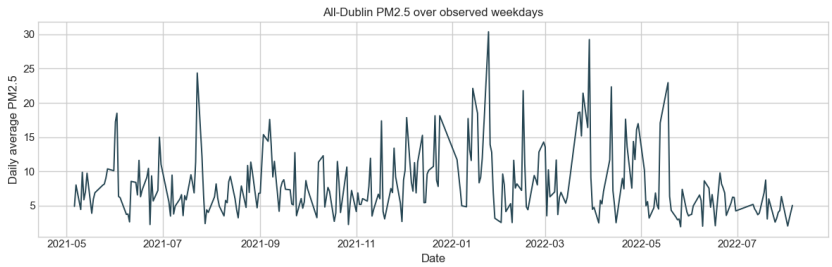
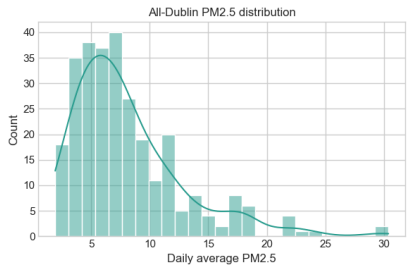
Objective and motivation

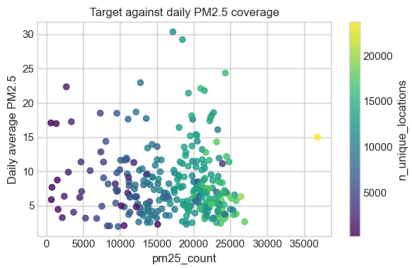
- Research Question: Can next-day air pollution in Dublin be predicted from recent pollution levels, weather and traffic conditions, and what does this imply for DCC's monitoring and intervention strategy?
- Target Variable of Interest: Particulate matter pollution, specifically PM2.5, particles with a size (diameter) of less than 2.5 micrometers
- PM2.5 has been identified as the main contributor to poor health from air pollution as the very small particles can penetrate deep into the respiratory system, contributing to more pollution-related deaths than ozone (Our World in Data 2025)
- We aimed to create an accurate forecasting model that can predict next day PM2.5 levels, to support Dublin City Council in their monitoring and prevention of air pollution (DCC Controlling Air Pollution)

Data Collected

- Google Air View Data: Street-view car pollutant measurements at 1 second intervals taken across 15 month period (May 2021- August 2022)
- Weather Data: Met Eirean Daily Historical Data at Dublin Airport
- Traffic Data: Traffic Volumes from SCATS Traffic Management System DCC








```
Weather data loaded.
Shape: (30740, 26)
```

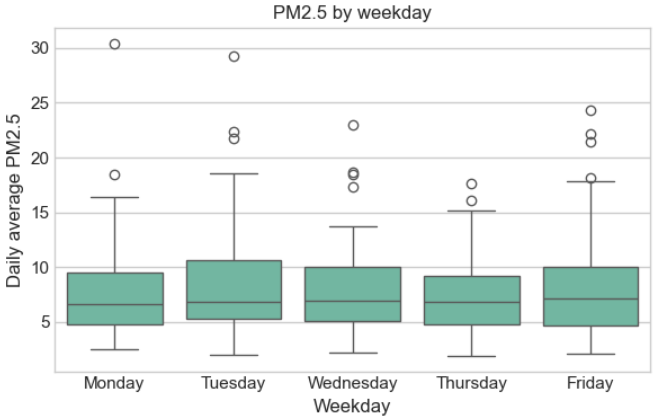
	date	ind	maxtp	ind.1	mintp	igmin	gmin	ind.2	rain	cbl	...	hg	sun	dos	g_rad	soil	pe	evap	smd_wd	smd_md	smd_pd
0	01-jan-1942	0	9.7	0	6.8	0	4.7	2	0.0	1020.3	...		0.0	0			1.1	1.4			
1	02-jan-1942	0	9.9	0	7.9	0	6.7	0	0.1	1016.2	...		0.0	0			0.7	0.9			
2	03-jan-1942	0	11.2	0	8.9	0	7.2	0	1.5	1006.8	...		0.1	0			0.5	0.6			
3	04-jan-1942	0	9.2	0	2.7	0	3.4	0	3.5	1001.5	...		0.6	0			0.6	0.7			
4	05-jan-1942	0	3.5	1	-0.8	0	0.0	0	0.6	1013.4	...		3.4	0			0.6	0.7			

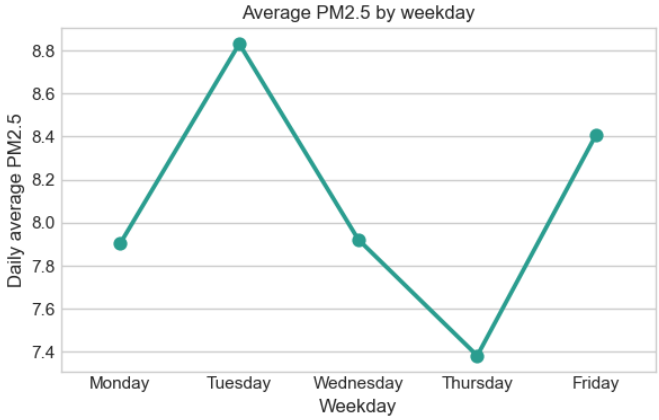
```
Traffic data loaded.
Shape: (3416, 6)
```

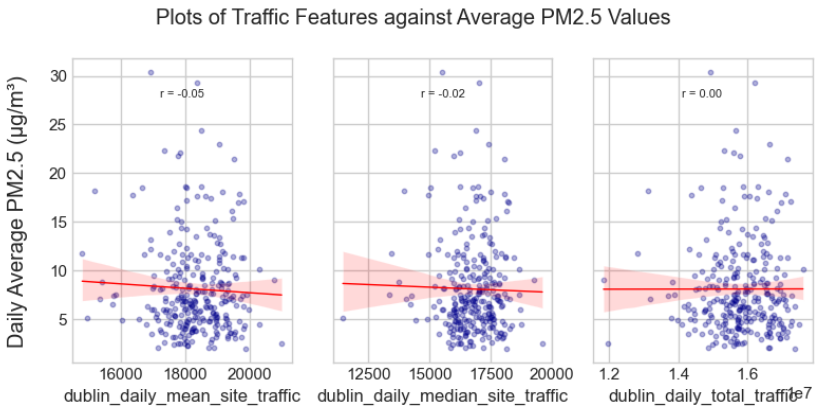
	date	Region	mean_daily_site_volume	median_daily_site_volume	total_daily_site_volume	n_sites
0	2021-05-01	CCITY	12098.288136	10559.5	1427598	118
1	2021-05-01	DCC1	17020.015152	14865.5	1123321	66
2	2021-05-01	IRE	16739.429319	14867.0	3197231	191
3	2021-05-01	IRE3	18313.857143	17985.0	512788	28
4	2021-05-01	NCITY	14283.241026	14182.0	2785232	195

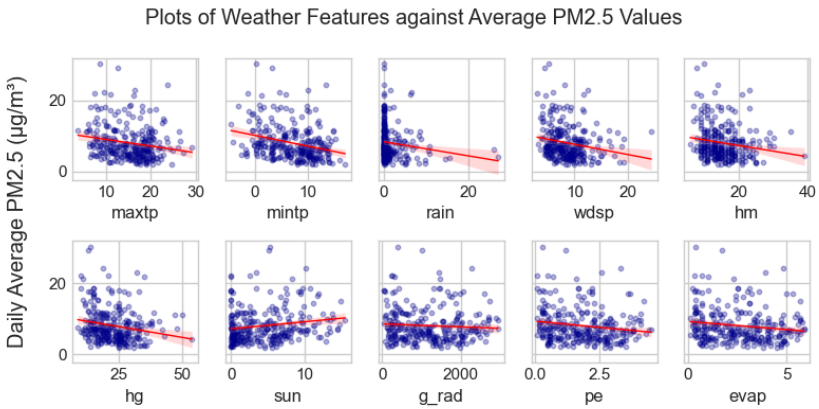
Pre-Processing

- Cleaning and Aggregation: matching time periods, removing missing/error PM2.5 values and aggregating air view and traffic data to daily Dublin target
- Merging: the processed datasets were combined by date, checking basic structure and saving final dataset for forecasting
- Visualisation: the data was explored revealing coverage restrictions, relationships between variables, spread of the data etc.



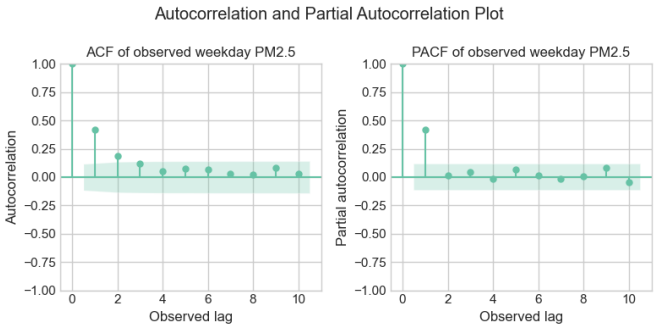


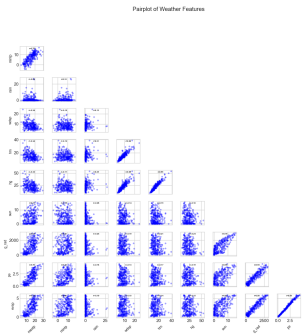




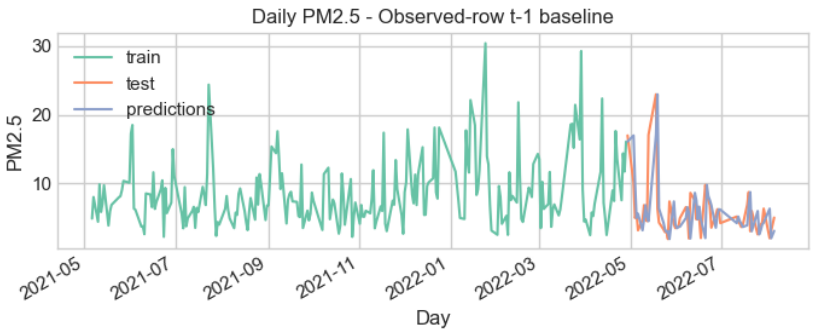
Method of Analysis

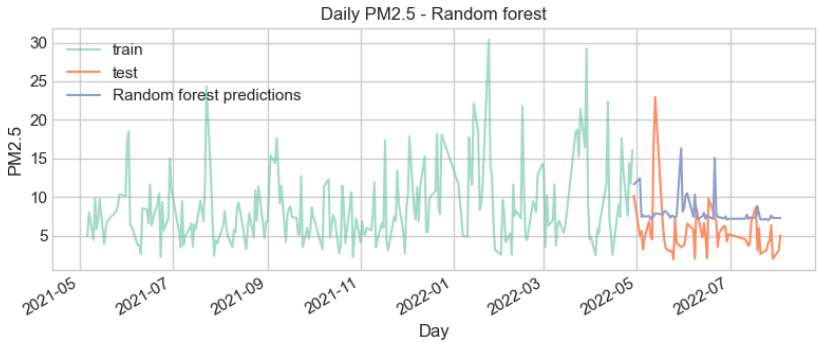
- Baseline Model: $\hat{y}_t = y_{t-1}$ simplest model predicting the observed test-day value based on the previous day
- Feature Selection: temporal features such as lag, window size and exogenous features such as weather and traffic
- Models Selection: Linear Regression, Same-week-day-average-forecaster, Direct one-step-ahead Ridge ARX, Random Forest

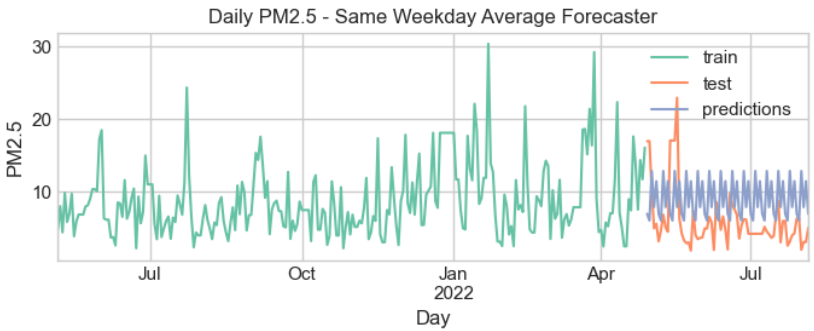


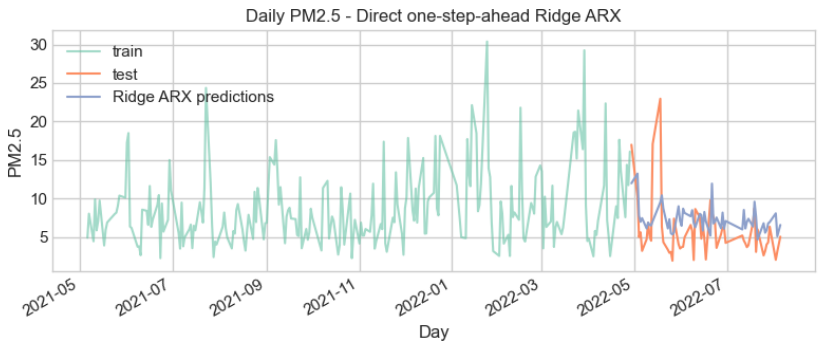


Results









- Baseline MSE: 14.96 RMSE: 3.86
- Linear Regression MSE: 15.52 RMSE: 3.94
- Same weekday average: MSE: 35.07 RMSE: 5.92
- Ridge ARX: MSE: 13.85 RMSE: 3.72
- Random Forest: RMSE: 4.62

Discussion and recommendation

- Final Model
  - Best full model: Ridge ARX
  - Only a small gain over the simple `t-1` baseline `RMSE 3.72 vs 3.87`
  - Main signal: yesterday's PM2.5; weather and traffic add only limited extra lift
- Limitations
  - Air View data is mobile, uneven, and weekday-only
  - Dublin Airport weather used as city-wide weather proxy
  - Traffic and weather are *lagged by design*: predict tomorrow using what is known today
- Recommendation to DCC
  - Use this as a light-touch next-day decision-support tool, not a standalone forecast
  - Prioritise better data coverage and monitoring design over more complex algorithms
  - Biggest gains likely from weekend coverage, stronger spatial balance, *forecast* weather, and *forward-looking* traffic inputs

Any Questions?